

L-CARNITINE

600 mg / 10 mL | 60 mg/mL | 10 VIALS | PRE-MIXED

Injectable L-Carnitine Solution -- Fat Oxidation & Energy Metabolism

Ready to Use | IM / IV Administration | Research Use Only

L-Carnitine (600 mg/10 mL) is a pre-mixed, ready-to-use injectable solution. No reconstitution required. Draw directly from the vial. Administered intramuscularly (IM) for standard dosing or intravenously (IV) for higher clinical/performance protocols. L-Carnitine is an endogenous amino acid derivative essential for mitochondrial fatty acid transport and energy metabolism.

Parameter	Specification
Active Compound	L-Carnitine (Levocarnitine) — pure L-isomer
Concentration	600 mg per 10 mL vial 60 mg/mL
Vial Volume	10 mL per vial 10 vials per kit 6,000 mg total
Format	Pre-mixed sterile aqueous solution — NO reconstitution required
Standard IM Dose	300-600 mg (0.5-1.0 mL) per injection
IV Dose	1,000-3,000 mg (16-50 mL) diluted in 0.9% NaCl for slow IV drip
Doses per Vial	20 doses at 0.5 mL (300mg) 10 doses at 1.0 mL (600mg)
Doses per Kit	200 doses at 300mg 100 doses at 600mg
Route	IM (deltoid or vastus lateralis) preferred IV slow drip (clinic only)
Needle (IM)	23-25G x 1 to 1.5 inch
Frequency	1-3x per week IM 1-2x per week IV
Best Timing	Pre-workout (30-60 min before) or morning fasted
Appearance	Clear, colourless to slightly yellow liquid. Discard if cloudy or particulate.
Storage	2-8 deg C refrigerated Protect from light Do not freeze

MECHANISM OF ACTION & RESEARCH BENEFITS

L-Carnitine is a quaternary ammonium compound synthesised endogenously from lysine and methionine. Its primary function is acting as an essential cofactor for the transport of long-chain fatty acids (LCFAs) across the inner mitochondrial membrane via the carnitine shuttle system, enabling beta-oxidation and ATP production. Injectable L-Carnitine bypasses intestinal absorption limitations, achieving significantly higher plasma concentrations than oral forms.

L-Carnitine Mechanism -- Mitochondrial Fatty Acid Transport



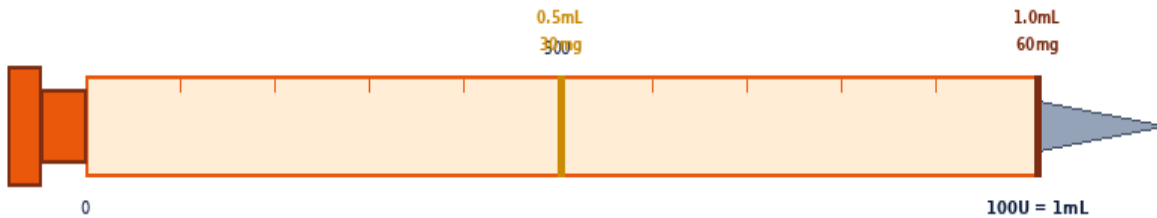
Pathway	Mechanism	Research Application
Fatty Acid Transport	Carnitine palmitoyltransferase I (CPT-I) converts acyl-CoA to acylcarnitine for transport across the inner mitochondrial membrane; CPT-II reconverts for beta-oxidation inside the matrix	Fat oxidation, body composition, weight management protocols
Energy Metabolism	Beta-oxidation of LCFAs produces acetyl-CoA feeding the citric acid cycle; generates NADH and FADH2 for electron transport chain and ATP synthesis	Athletic performance, fatigue reduction, mitochondrial support
Cardiac Support	Heart muscle relies heavily on fatty acid oxidation for ~60-70% of energy; carnitine deficiency impairs cardiac contractility; supplementation supports myocardial energy production	Cardiac research, heart failure models, cardiometabolic studies
Insulin Sensitivity	Modulates glucose and fatty acid metabolism competition; reduces intramyocellular lipid accumulation; improves glucose utilisation in skeletal muscle	Metabolic syndrome research, Type 2 diabetes models
Neuroprotection	Acetyl-L-carnitine (ALCAR) form crosses blood-brain barrier; supports acetylcholine synthesis; mitochondrial energy in neurons; antioxidant activity via thiol groups	Cognitive function, peripheral neuropathy, neurological research

Key Research Benefits

Benefit	Evidence & Context
Enhanced Fat Oxidation	Injectable L-Carnitine achieves plasma levels 3-5x higher than oral supplementation, maximising fatty acid transport into mitochondria for energy production.
Athletic Performance & Recovery	Research shows reduced lactate accumulation, improved VO2 max, faster recovery, and reduced exercise-induced muscle damage markers (CK, LDH).
Carnitine Deficiency Correction	Primary and secondary carnitine deficiencies (genetic, dialysis-related, vegetarian diets) respond rapidly to injectable L-Carnitine.
Metabolic & Cardiac Research	Extensive clinical research base in cardiovascular disease, peripheral arterial disease, and metabolic syndrome; well-established safety and efficacy profile.

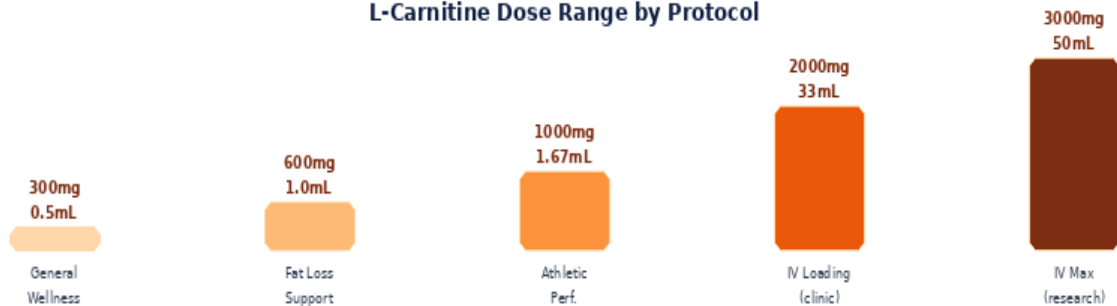
DOSE GUIDE & INJECTION PROCEDURE

No reconstitution needed. Draw directly from vial. Warm to room temperature 10-15 minutes before injecting. For IV use, dilute the required volume in 100-250 mL of 0.9% sodium chloride and administer as a slow drip over 30-60 minutes.



Standard 1mL syringe for IM | 10mL vial for IV administration

L-Carnitine Dose Range by Protocol



Protocol	Dose	Volume from Vial	Route	Frequency	Timing
Wellness / Recovery	300 mg	0.5 mL	IM	3x / week	Pre-workout or morning
Fat Loss / Body Comp.	600 mg	1.0 mL	IM	3x / week	30-60 min pre-workout
Athletic Performance	1,000 mg	1.67 mL	IM or IV	2-3x / week	30-60 min pre-event
IV Loading	2,000 mg	33 mL + 0.9% NaCl	IV drip only	1-2x / week	Clinic setting; 30-60 min infusion
Deficiency Repletion	3,000 mg	50 mL + 0.9% NaCl	IV drip only	Daily x 5 days then weekly	Under medical supervision only

Step-by-Step IM Injection Procedure

Step	Action	Detail
1	Warm the vial	Remove from fridge 10-15 minutes before use. Cold solution increases injection discomfort.
2	Wash hands	Wash with soap and warm water for 20 seconds. Dry completely.
3	Select needle & syringe	IM: 23-25G x 1 to 1.5 inch syringe (1 mL or 3 mL). IV: standard IV cannula or butterfly needle.
4	Clean vial stopper	Wipe rubber stopper with a fresh alcohol swab. Air-dry 10 seconds.

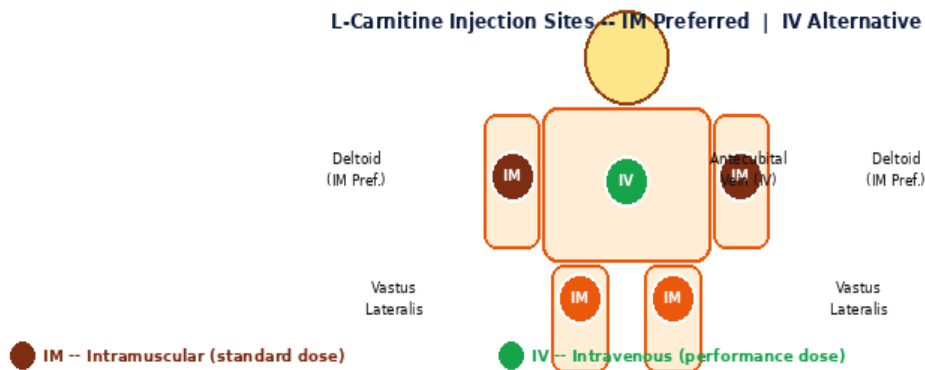
5	Draw the dose	Draw the prescribed volume. Remove all air bubbles before injection.
6	Clean injection site	Wipe the deltoid or thigh with an alcohol swab. Allow 10-15 sec to dry.
7	Inject	Insert at 90 degrees with a smooth dart motion. Aspirate briefly -- if blood draws back, withdraw and select new site. Inject slowly over 5-10 seconds.
8	Withdraw and press	Remove at same angle. Press with dry swab 5-10 seconds. Do NOT rub. Mild warmth or local flush is normal.
9	Dispose safely	Needle immediately into sharps container. Never recap, reuse, or leave exposed.

INJECTION SITES, ROTATION & SCHEDULE

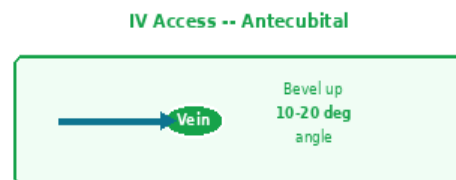
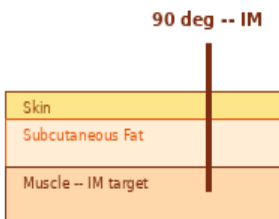
IM injection into the deltoid is the preferred route for standard dosing — fastest absorption, easiest self-administration. The **vastus lateralis (outer thigh)** is a good secondary site. For **IV administration** (1,000-3,000 mg), dilute in 0.9% NaCl and infuse slowly over 30-60 minutes in a clinical setting. Rotate IM sites every session to prevent localised irritation.

Site	Route	Needle	Volume Limit	Notes
Deltoid (upper outer arm)	IM (Preferred)	23-25G 1 to 1.5 in	Up to 2 mL	Alternate left/right arm. Fastest absorption.
Vastus Lateralis (outer thigh)	IM (Alt.)	23-25G 1 to 1.5 in	Up to 5 mL	Middle outer thigh. Good for larger volumes.
Antecubital vein (inner elbow)	IV only	21-23G IV cannula	Diluted in 100-250 mL NaCl	Clinical setting only. Slow drip 30-60 min.

L-Carnitine Injection Sites -- IM Preferred | IV Alternative



IM (90 deg) and IV Access Technique



Deltoid Rotation -- 4 Points per Arm, Alternate Arms Each Session



Left Deltoid



Allow 1 week rest per site before re-injecting same point



Right Deltoid

Recommended Schedule -- 3x Per Week (Mon / Wed / Fri)



Inject pre-workout or morning fasted for optimal fat oxidation effect

SIDE EFFECTS, STORAGE & GOLDEN RULES

Side Effect	Likelihood	Management
Injection site pain / warmth	Common	Warm vial before injecting. Inject slowly. Rotate sites. Resolves quickly.
Fishy body odour (TMAO)	Common	Gut bacteria convert some carnitine to trimethylamine. Reduces with time. Contraindicated in trimethylaminuria.
Nausea (rare at IM doses)	Occasional	Inject with a light meal if IM. IV infusion nausea is rate-related -- slow the drip rate.
Flushing / warmth (IV)	Occasional (IV)	Rate-related with IV. Slow infusion rate to 30-60 min minimum.
Mild diarrhoea (high doses)	Occasional	More common at oral doses. IM/IV bypasses gut. Reduce dose if persistent.
Restlessness / insomnia	Occasional	Energy-stimulating effect of high doses. Inject morning or pre-workout only.
Seizure risk (end-stage renal disease)	Rare	High IV doses in ESRD patients may lower seizure threshold. Medical supervision required.
Allergic reaction	Rare	Stop immediately. Seek medical advice. Do not rechallenge.

STOP & SEEK MEDICAL ADVICE: Severe allergic reaction: hives, facial swelling, difficulty breathing

STOP & SEEK MEDICAL ADVICE: Seizure or loss of consciousness (especially in renal-impaired individuals)

STOP & SEEK MEDICAL ADVICE: Severe chest pain or cardiac arrhythmia

STOP & SEEK MEDICAL ADVICE: Signs of injection site infection: spreading redness, increasing warmth, pus, fever

Contraindications: Known allergy to L-Carnitine or any component, trimethylaminuria (fish odour syndrome), end-stage renal disease without medical supervision (seizure risk at high IV doses), pregnancy or lactation (consult physician). Not for use under 18.

Storage Guidelines

State	Temperature	Duration	Notes
Unopened vial	2-8 deg C	Per expiry	Upright. Away from light. Do not freeze.
Opened / in-use	2-8 deg C	28 days	Cap securely after each draw. Inspect before use.
Prepared IV bag	2-8 deg C or room temp	Use within 4 hr	Diluted L-Carnitine in NaCl should be used promptly.
Short-term transport	Below 25 deg C	Max 72 hr	Insulated bag with ice packs. No direct ice contact.

Golden Rules — L-Carnitine Injectable

1. Warm the vial to room temperature 10-15 minutes before injecting -- cold solution stings.
2. Draw directly from the vial -- NO reconstitution required.
3. IM injection into the deltoid is the preferred route for standard 300-600mg dosing.
4. For IV doses above 1,000mg -- dilute in 100-250mL of 0.9% NaCl and infuse slowly over 30-60 minutes.
5. Never inject IV doses as a rapid bolus -- infuse slowly to avoid nausea and flushing.

6. Aspirate before IM injection -- withdraw and change site if blood draws back.
7. Inject 30-60 minutes pre-workout or in the morning for optimal fat oxidation benefit.
8. Rotate IM sites every session. Allow at least 1 week rest per injection point.
9. Discard any opened vial after 28 days. Prepared IV bags must be used within 4 hours.
10. Contraindicated in trimethylaminuria and end-stage renal disease without medical supervision.

DISCLAIMER: This document is produced by for informational and research reference purposes only. This guide does not constitute medical advice. IV administration of L-Carnitine must only be performed by a qualified medical professional. assumes no liability for misuse of this product.

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